

Code-Layout, Readability and Reusability

Date	07 July 2026
Team ID	SWTID-2026-2168
Project Name	EduGenie: Google Gemini-Powered Learning Assistant
Maximum Marks	5 Marks

Code-Layout, Readability and Reusability:

This document evaluates the quality of the codebase in terms of its structure, readability, and reusability. Well-organized code with proper naming conventions, comments, and modular design ensures that the project is maintainable and can be extended or reused in future development.

Code Layout Checklist:

S.No	Code Quality Parameter	Description	Followed (Yes / No / Partial)	Remarks
1	Consistent Indentation	Uniform spacing/tabs used throughout the code	Yes	Uniform formatting
2	Proper File Structure	Files and folders are logically organized	Yes	Organized folders
3	Meaningful Variable Names	Variables reflect their purpose clearly	Yes	Clear naming
4	Function / Method Names	Functions are descriptively named	Yes	Descriptive methods
5	Code Comments	Inline and block comments explain logic	Yes	Basic comments added
6	Modular Design	Code is split into reusable functions/module	Yes	Separate frontend/backend
7	No Redundant Code	Duplicate or unused code is removed	Yes	Duplicate code removed
8	Error Handling	Exceptions and errors are handled gracefully	Yes	Exception handling used

Reusable Components / Modules:

S.No	Component / Module Name	Language / Technolog	yWhere Reused	Reusability Level (High / Medium / Low)
1	API Services	Python / FastAPI	Backend APIs	High
2	AI Prompt Module	Python	AI Features	High
3	UI Components	HTML, Tailwind CSS	Frontend	High
4	JavaScript Functions	JavaScript	Multiple Pages	Medium
5	Gemini API Integration	Python	AI Services	High

Overall Code Quality Assessment:

Aspect	Rating (1-5)	Comments
Code Layout & Structure	5	Well organized
Readability	5	Easy to understand
Reusability	5	Modular design
Documentation / Comments	4	Basic comments included
Overall Score	4.8	Good quality code